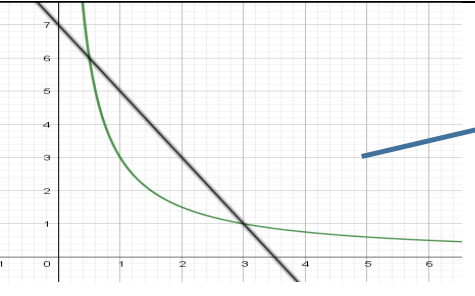
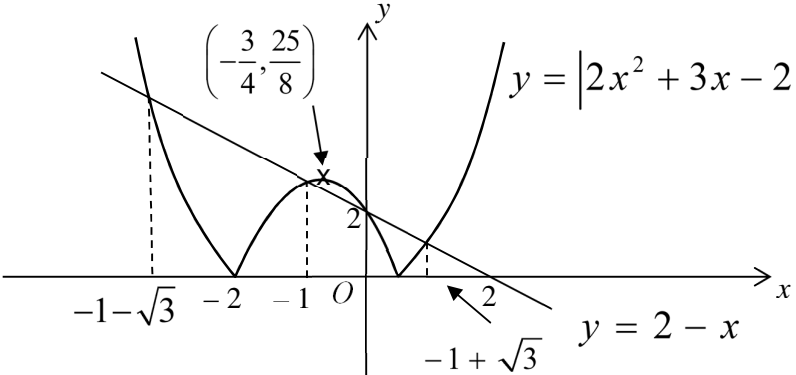


2018 H2 Math 9758 Paper 1 Solution

	Solution	Comments
1(i)	$y = \frac{\ln x}{x}$ $\frac{dy}{dx} = \frac{x\left(\frac{1}{x}\right) - \ln x(1)}{x^2}$ $= \frac{1 - \ln x}{x^2}$	Apply Quotient Rule
1(ii)	$\int_1^e \frac{\ln x}{x^2} dx$ $= -\int_1^e \frac{-\ln x}{x^2} dx$ $= -\int_1^e \frac{1 - \ln x - 1}{x^2} dx$ $= -\int_1^e \frac{1 - \ln x}{x^2} - \frac{1}{x^2} dx$ $= -\int_1^e \frac{1 - \ln x}{x^2} - x^{-2} dx$ $= -\left[\frac{\ln x}{x} - \frac{x^{-1}}{-1} \right]_1^e$ $= -\left[\frac{\ln x}{x} + \frac{1}{x} \right]_1^e$ $= -\left[\left(\frac{1}{e} + \frac{1}{e} \right) - (0 + 1) \right]$ $= 1 - \frac{2}{e}$	Hence Question: From part (i), we have: $\int_1^e \frac{1 - \ln x}{x^2} dx = \frac{\ln x}{x} + C$ So we try to manipulate for such as expression to appear in our integral. Step 1: Multiply by 2 negative signs Step 2: Add 1 and Subtract 1 in the numerator Step 3: Split the numerator
2(i)	$y = \frac{3}{x} \quad \text{--- (1)}$ $y + 2x = 7 \quad \text{--- (2)}$ Sub (1) into (2) to find intersection points: $\frac{3}{x} + 2x = 7$ $3 + 2x^2 = 7x$ $2x^2 - 7x + 3 = 0$ $(2x - 1)(x - 3) = 0$ $\therefore$ The x-coordinates of A and B are $\frac{1}{2}$ and 3.	Note: You are to show every single step including factorising as calculator is NOT allowed in solving this question. You can of course use your calculator to check your factorisation. Be smart.

(ii)	 $ \begin{aligned} \text{Volume} &= \pi \int_{0.5}^3 (7-2x)^2 dx - \pi \int_{0.5}^3 \frac{9}{x^2} dx \\ &= \pi \int_{0.5}^3 (7-2x)^2 - 9x^{-2} dx \\ &= \pi \left[\frac{1}{3} \frac{(7-2x)^3}{-2} - \frac{9x^{-1}}{-1} \right]_{0.5}^3 \\ &= \pi \left[-\frac{1}{6} (7-2x)^3 + \frac{9}{x} \right]_{0.5}^3 \\ &= \pi \left(-\frac{1}{6} + 3 + 36 - 18 \right) \\ &= \frac{125}{6} \pi \end{aligned} $	Although you cannot use calculator to solve the question, you can use it to give a rough sketch of the region bounded by the line and curve. This helps in formulating the volume. From the sketch, we can rotate the line about the x-axis minus rotate the curve about the x-axis. Line: $y + 2x = 7$ $\Rightarrow y = 7 - 2x$ $\Rightarrow y^2 = (7 - 2x)^2$ Curve: $y = \frac{3}{x}$ $\Rightarrow y^2 = \frac{9}{x^2}$
3(i)	$y = ux^2 \quad \text{---(1)}$ $\frac{dy}{dx} = u(2x) + x^2 \frac{du}{dx} \quad \text{---(2)}$ Given $x \frac{dy}{dx} = 2y - 6 \quad \text{---(3)}$ Sub (1) and (2) into (3) $x \left(x^2 \frac{du}{dx} + 2ux \right) = 2ux^2 - 6$ $x^3 \frac{du}{dx} = -6$ $\frac{du}{dx} = \frac{-6}{x^3} \quad (\text{since } x \neq 0)$ $\left(\text{checked using } y = ux^2 \text{ and } x \frac{dy}{dx} = 2y - 6 \right)$ $\therefore f(x) = \frac{-6}{x^3}$	Step 1: Write down given substitution and label as (1). Step 2: Differentiate (1) with respect to the variable as shown in question and label it as (2). Step 3: Write down given DE and label as (3). Step 4: Sub (1) and (2) into (3) and simplify.

(ii)	$\frac{du}{dx} = \frac{-6}{x^3}$ $\int 1 du = \int -6x^{-3} dx$ $u = \frac{-6x^{-2}}{-2} + C$ $u = \frac{3}{x^2} + C$ Sub (1) into the above equation: $\frac{y}{x^2} = \frac{3}{x^2} + C$ $y = 3 + Cx^2$ When $x = 1$ and $y = 2$, $C = -1$. $\therefore y = 3 - x^2$	Step 5: Solve the simplified DE.
4(i)	$ 2x^2 + 3x - 2 = 2 - x$ $2x^2 + 3x - 2 = 2 - x \quad \text{or} \quad 2x^2 + 3x - 2 = -(2 - x)$ $2x^2 + 4x - 4 = 0 \quad \text{or} \quad 2x^2 + 2x = 0$ $x^2 + 2x - 2 = 0 \quad \text{or} \quad x^2 + x = 0$ $x = \frac{-2 \pm \sqrt{2^2 - 4(1)(-2)}}{2(1)} \quad \text{or} \quad x(x+1) = 0$ $x = -1 \pm \sqrt{3} \quad \text{or} \quad x = 0 \quad \text{or} \quad x = -1$ $\therefore$ The roots are $-1 - \sqrt{3}$, -1, 0 and $-1 + \sqrt{3}$.	$x = a$ $\Rightarrow x = a \quad \text{or} \quad x = -a$
(ii)	 From the graph, $-1 - \sqrt{3} < x < -1 \quad \text{or} \quad 0 < x < -1 + \sqrt{3}.$	To solve $2x^2 + 3x - 2 < 2 - x$, We need to find the values of x such that the graph of $y = 2x^2 + 3x - 2$ is below the graph of $y = 2 - x$.

5	<u>Method 1:</u> Since $ff = g$, it means that ff exists. This means that $R_f \subseteq D_f$ $D_f = \mathbb{R} \setminus \{-b\}$ $f(x) = \frac{x+a}{x+b} = 1 + \frac{a-b}{x+b}$ Horizontal asymptote at $y = 1$ Vertical asymptote at $x = -b$ $R_f = \mathbb{R} \setminus \{1\}$ Since $R_f \subseteq D_f \Rightarrow \mathbb{R} \setminus \{1\} \subseteq \mathbb{R} \setminus \{-b\}$ $\therefore b = -1$ <u>Method 2:</u> Since $ff(x) = g(x) = x$, then we have $f^{-1}ff(x) = f^{-1}(x)$ $f(x) = f^{-1}(x)$ Let $y = f(x)$. $y = \frac{x+a}{x+b}$ $xy + by = x + a$ $xy - x = -by + a$ $x = \frac{-by + a}{y - 1}$ $f^{-1}(x) = \frac{-bx + a}{x - 1}$ Since $f(x) = f^{-1}(x)$, we have $\frac{x+a}{x+b} = \frac{-bx+a}{x-1} \Rightarrow b = -1$ <u>Method 3:</u> $ff(x) = g(x) = x$ $\frac{\frac{x+a}{x+b} + a}{\frac{x+a}{x+b} + b} = x$ $\frac{x(a+1) + a(b+1)}{x(b+1) + a + b^2} = x$ $x(a+1) + a(b+1) = x(x(b+1) + a + b^2)$ $(b+1)x^2 + (a+b^2 - (a+1))x - a(b+1) = 0$ for $x \in \mathbb{R} \setminus \{-b\}$ Comparing coefficient of x^2, $b = -1$	
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	For Method 1 and 3: $f(f(x)) = g(x) = x$ $f(f(x)) = x$ $f^{-1}(f(f(x))) = f^{-1}(x) \quad (\text{since } f^{-1} \text{ exists})$ $f(x) = f^{-1}(x)$ $f^{-1}(x) = \frac{x+a}{x-1}$ For Method 2: $f^{-1}(x) = f(x) = \frac{x+a}{x-1}$	Takes f^{-1} on both sides.
6(i)	$\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$ $\mathbf{a} \times 3\mathbf{b} - 2\mathbf{a} \times \mathbf{c} = \mathbf{0}$ $\mathbf{a} \times (3\mathbf{b} - 2\mathbf{c}) = \mathbf{0}$ $3\mathbf{b} - 2\mathbf{c} = \mathbf{0} \quad \text{or} \quad 3\mathbf{b} - 2\mathbf{c} \text{ is parallel to } \mathbf{a}$ $\therefore 3\mathbf{b} - 2\mathbf{c} = \lambda \mathbf{a}, \lambda \in \mathbb{R}.$	$\mathbf{0}$ is the zero vector. Learning point: Vector (Cross) product gives a vector.
6(ii)	We are given that $\mathbf{a}$ and $\mathbf{c}$ are unit vectors, that the modulus of $\mathbf{b}$ is 4 and that the angle between $\mathbf{b}$ and $\mathbf{c}$ is 60°. Meaning: $\mathbf{a} = 1, \mathbf{c} = 1, \mathbf{b} = 4$ and $\mathbf{b} \cdot \mathbf{c} = \mathbf{b} \mathbf{c} \cos 60^\circ$ $\therefore 3\mathbf{b} - 2\mathbf{c} = \lambda \mathbf{a}$ $(3\mathbf{b} - 2\mathbf{c}) \cdot (3\mathbf{b} - 2\mathbf{c}) = \lambda \mathbf{a} \cdot \lambda \mathbf{a}$ $9\mathbf{b} \cdot \mathbf{b} - 6\mathbf{b} \cdot \mathbf{c} - 6\mathbf{c} \cdot \mathbf{b} + 4\mathbf{c} \cdot \mathbf{c} = \lambda^2 \mathbf{a} \cdot \mathbf{a}$ $9 \mathbf{b} ^2 - 12\mathbf{b} \cdot \mathbf{c} + 4 \mathbf{c} ^2 = \lambda^2 \mathbf{a} ^2$ $9 \mathbf{b} ^2 - 12 \mathbf{b} \mathbf{c} \cos 60^\circ + 4 \mathbf{c} ^2 = \lambda^2 \mathbf{a} ^2$ $9(4)^2 - 12(4)(1)\cos 60^\circ + 4(1)^2 = \lambda^2 (1)^2$ $144 - 48\cos 60^\circ + 4 = \lambda^2$ $\lambda^2 = 124$ $\lambda = \pm 2\sqrt{31}$ $\therefore \lambda = -2\sqrt{31} \text{ or } \lambda = 2\sqrt{31}.$	To use magnitudes of vectors (as given by question), we need to consider the use of scalar product as $\mathbf{u} \cdot \mathbf{u} = \mathbf{u} ^2$. Using part (i) result, we consider $(3\mathbf{b} - 2\mathbf{c}) \cdot (3\mathbf{b} - 2\mathbf{c}) = \lambda \mathbf{a} \cdot \lambda \mathbf{a}$ Learning Points: $\mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{b}$ (commutative) BUT (Bonus for you) $\mathbf{b} \times \mathbf{c} = -\mathbf{c} \times \mathbf{b}$ Switching order for cross product needs a negative sign.

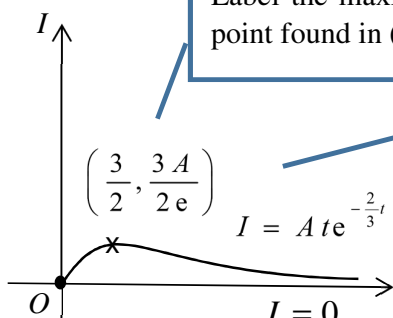
7(i)	$\frac{x^2 - 4y^2}{x^2 + xy^2} = \frac{1}{2}$ $2x^2 - 8y^2 = x^2 + xy^2$ $x^2 - 8y^2 = xy^2$ $2x - 16y \frac{dy}{dx} = x \left(2y \frac{dy}{dx} \right) + y^2 (1)$ $2xy \frac{dy}{dx} + 16y \frac{dy}{dx} = 2x - y^2$ $\frac{dy}{dx} (2xy + 16y) = 2x - y^2$ $\frac{dy}{dx} = \frac{2x - y^2}{2xy + 16y} \quad (\text{Shown})$	Not easy to differentiate in this form using quotient rule. Cross Multiply the fractions so we can apply product rule instead. Show every single step.
(ii)	When $x = 1$, $\frac{1 - 4y^2}{1 + y^2} = \frac{1}{2}$ $2 - 8y^2 = 1 + y^2$ $9y^2 = 1$ $y = \pm \frac{1}{3}$ When $y = -\frac{1}{3}$, $\frac{dy}{dx} = \frac{2 - \left(\frac{1}{3}\right)^2}{-2\left(\frac{1}{3}\right) - 16\left(\frac{1}{3}\right)}$ $= -\frac{17}{54}$ Therefore, the equation of tangents at $x = 1$ are $y - \left(-\frac{1}{3}\right) = -\frac{17}{54}(x - 1) \quad \text{--- (1)} \quad \text{and}$ $y - \frac{1}{3} = \frac{17}{54}(x - 1) \quad \text{--- (2)}.$ Equating (1) and (2), $y - \left(-\frac{1}{3}\right) = -\left(y - \frac{1}{3}\right)$ $y + \frac{1}{3} = -y + \frac{1}{3}$ $y = 0$	Equation of tangent is given by $y - y_1 = m(x - x_1)$ So we need to find 1) x_1 (given) 2) y_1 3) $m = \frac{dy}{dx}$ evaluated at (x_1, y_1) To find intersection between two lines (curves), we have to solve simultaneous equations.

	Sub $y = 0$ into (1) $\frac{1}{3} = -\frac{17}{54}(x-1)$ $x-1 = -\frac{18}{17}$ $x = -\frac{1}{17}$ $\therefore$ coordinates of N is $\left(-\frac{1}{17}, 0\right)$.										
8(i)	$u_{n+1} = 2u_n + An, n \geq 1$ Sub $n = 1, u_1 = 5$ $u_2 = 2u_1 + A(1)$ $15 = 2(5) + A(1)$ $15 = 10 + A$ $A = 5$ Sub $n = 2, u_2 = 15$ $u_3 = 2u_2 + 5(2)$ $= 2(15) + 10$ $= 40$										
(ii)	$u_n = a(2^n) + bn + c$ <table border="0"> <tr> <td>Sub $n = 1, u_1 = 5$</td> <td>Sub $n = 2, u_2 = 15$</td> <td>Sub $n = 3, u_3 = 40$</td> </tr> <tr> <td>$u_1 = a(2^1) + b + c$</td> <td>$u_2 = a(2^2) + 2b + c$</td> <td>$u_3 = a(2^3) + 3b + c$</td> </tr> <tr> <td>$2a + b + c = 5$</td> <td>$4a + 2b + c = 15$</td> <td>$8a + 3b + c = 40$</td> </tr> </table> Using GC, $\therefore a = 7.5, b = -5$ and $c = -5$.	Sub $n = 1, u_1 = 5$	Sub $n = 2, u_2 = 15$	Sub $n = 3, u_3 = 40$	$u_1 = a(2^1) + b + c$	$u_2 = a(2^2) + 2b + c$	$u_3 = a(2^3) + 3b + c$	$2a + b + c = 5$	$4a + 2b + c = 15$	$8a + 3b + c = 40$	
Sub $n = 1, u_1 = 5$	Sub $n = 2, u_2 = 15$	Sub $n = 3, u_3 = 40$									
$u_1 = a(2^1) + b + c$	$u_2 = a(2^2) + 2b + c$	$u_3 = a(2^3) + 3b + c$									
$2a + b + c = 5$	$4a + 2b + c = 15$	$8a + 3b + c = 40$									
(iii)	$u_n = 7.5(2^n) - 5n - 5$ $u_r = 7.5(2^r) - 5r - 5$	Since we need to find $\sum_{r=1}^n u_r$, we first need u_r. Given $u_n = 7.5(2^n) - 5n - 5$, (Replace n by r)									

	$\sum_{r=1}^n u_r$ $= \sum_{r=1}^n (7.5(2^r) - 5r - 5)$ $= 7.5 \sum_{r=1}^n 2^r - 5 \sum_{r=1}^n (r+1)$ $= 7.5(2^1 + 2^2 + 2^3 + \dots + 2^n) - 5(2 + 3 + 4 + \dots + (n+1))$ $= 7.5 \left(\frac{2}{1-2} (1-2^n) \right) - 5 \left(\frac{n}{2} (2 + (n+1)) \right)$ $= 15(2^n - 1) - \frac{5n}{2}(n+3)$	Note: r is the dummy variable. 2^r has the variable in power so it is a Geometric Series. $(r+1)$ is linear in terms of r so it is an Arithmetic Series. So we need to split the summation to find the sum separately.	
		Sum of GP: $\frac{\text{first term}}{1-r} (1 - r^{\text{no. of terms}}), \quad r = \text{common ratio}$ Sum of AP: $\frac{\text{no. of terms}}{2} (\text{1st} + \text{last})$	
9(i)	$x = 2\theta - \sin(2\theta)$ $\frac{dx}{d\theta} = 2 - 2\cos(2\theta)$ $\frac{dy}{dx} = \frac{4\sin\theta\cos\theta}{2 - 2\cos(2\theta)}$ $= \frac{4\sin\theta\cos\theta}{2 - 2(1 - 2\sin^2\theta)}$ $= \frac{4\sin\theta\cos\theta}{2 - 2 + 4\sin^2\theta}$ $= \frac{4\sin\theta\cos\theta}{4\sin^2\theta}$ $= \frac{\cos\theta}{\sin\theta}$ $= \cot\theta \quad (\text{Shown})$	$y = 2\sin^2\theta$ $\frac{dy}{d\theta} = 4\sin\theta\cos\theta$ To get $\cot\theta = \frac{\cos\theta}{\sin\theta}$, we cannot have $\cos(2\theta)$ as the angle is not the same. Thus, we need to apply double angle formula: $\cos(2\theta) = \cos^2\theta - \sin^2\theta$ $= 2\cos^2\theta - 1$ $= 1 - 2\sin^2\theta$ We choose $\cos(2\theta) = 1 - 2\sin^2\theta$ as it helps to eliminate the constant in the denominator and it helps to simplify the expression to $\frac{\cos\theta}{\sin\theta}$.	

(ii)	At $\theta = \alpha$, $x = 2\alpha - \sin 2\alpha$ $y = 2\sin^2 \alpha$ $\frac{dy}{dx} = \cot \alpha \Rightarrow -\frac{1}{m} = -\frac{1}{\cot \alpha} = -\tan \alpha$ Equation of Normal to curve at $\theta = \alpha$: $y - 2\sin^2 \alpha = -\tan \alpha (x - (2\alpha - \sin 2\alpha))$ At Point A, cutting the x-axis, $y = 0$, $0 - 2\sin^2 \alpha = -\tan \alpha (x - (2\alpha - \sin 2\alpha))$ $0 - 2\sin^2 \alpha = -\frac{\sin \alpha}{\cos \alpha} (x - (2\alpha - \sin 2\alpha))$ $x - (2\alpha - \sin 2\alpha) = 2\sin \alpha \cos \alpha$ $x - 2\alpha + \sin 2\alpha = \sin 2\alpha$ $x = 2\alpha$ $\therefore k = 2$	Equation of normal is given by $y - y_1 = -\frac{1}{m}(x - x_1)$ So we need to find 1) x_1 2) y_1 3) $m = \frac{dy}{dx}$ evaluated at (x_1, y_1)
(iii)	From part (i), $\frac{dx}{d\theta} = 2 - 2\cos(2\theta) \quad \frac{dy}{d\theta} = 4\sin \theta \cos \theta$ And note that given $0 \leq \theta \leq \pi$ $\int_{\beta}^{\gamma} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta.$ $= \int_0^{\pi} \sqrt{(2 - 2\cos(2\theta))^2 + (4\sin \theta \cos \theta)^2} d\theta$ $= \int_0^{\pi} \sqrt{4 - 8\cos(2\theta) + 4\cos^2(2\theta) + (2\sin(2\theta))^2} d\theta$ $= \int_0^{\pi} \sqrt{4 - 8\cos(2\theta) + 4\cos^2(2\theta) + 4\sin^2(2\theta)} d\theta$ $= \int_0^{\pi} \sqrt{4 - 8\cos(2\theta) + 4(\cos^2(2\theta) + \sin^2(2\theta))} d\theta$ $= \int_0^{\pi} \sqrt{4 - 8\cos(2\theta) + 4(1)} d\theta$ $= \int_0^{\pi} \sqrt{8 - 8\cos(2\theta)} d\theta$ $= \int_0^{\pi} \sqrt{16\sin^2 \theta} d\theta$ $= \int_0^{\pi} 4\sin \theta d\theta \quad (\text{since } 0 \leq \theta \leq \pi)$ $= [-4\cos \theta]_0^{\pi}$ $= 8$	Note: $\sin(2\theta) = 2\sin \theta \cos \theta$ We need to ensure the angles are the same to simplify the trigonometric expression. We need to apply double angle formula: $\begin{aligned} \cos(2\theta) &= \cos^2 \theta - \sin^2 \theta \\ &= 2\cos^2 \theta - 1 \\ &= 1 - 2\sin^2 \theta \end{aligned}$ We choose $\cos(2\theta) = 1 - 2\sin^2 \theta$ as it helps to eliminate the constant and make the expression easy for taking square root.

10(i)	Note: 1) L, R and C are constants. 2) I, t and q are variables. 3) V is not stated so take it as variable first $L \frac{dI}{dt} + RI + \frac{q}{C} = V$ Differentiating with respect to t, $L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{1}{C} \frac{dq}{dt} = \frac{dV}{dt}$ $L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = \frac{dV}{dt} \quad \left(\because I = \frac{dq}{dt} \right)$ By comparing with given DE, $\frac{dV}{dt} = 0$ This means that V is a constant. $L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = 0 \quad (\text{Shown})$	
(ii)	$I = Ate^{\frac{Rt}{2L}}$ $\frac{dI}{dt} = Ae^{\frac{Rt}{2L}} + A \left(-\frac{R}{2L} \right) te^{\frac{Rt}{2L}}$ $= Ae^{\frac{Rt}{2L}} - \frac{R}{2L} I \quad \text{----- (1)}$ $\frac{d^2 I}{dt^2} = A \left(-\frac{R}{2L} \right) e^{\frac{Rt}{2L}} - \frac{R}{2L} \frac{dI}{dt}$ $= -\frac{R}{2L} \left(\frac{dI}{dt} + \frac{R}{2L} I \right) - \frac{R}{2L} \frac{dI}{dt} \quad (\text{from (1)})$ $\frac{d^2 I}{dt^2} = -\frac{R}{L} \frac{dI}{dt} - \left(\frac{R}{2L} \right)^2 I$ $\frac{d^2 I}{dt^2} + \frac{R}{L} \frac{dI}{dt} + \left(\frac{R}{2L} \right)^2 I = 0$ $L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{R^2}{4L} I = 0$ By comparing coefficients of I with $L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = 0$, $\frac{1}{C} = \frac{R^2}{4L}$ $C = \frac{4L}{R^2} \quad (\text{Shown})$ Only I and t are variables. The rest of the letters are constant. From (1), $\frac{dI}{dt} = Ae^{\frac{Rt}{2L}} - \frac{R}{2L} I$ $Ae^{\frac{Rt}{2L}} = \frac{dI}{dt} + \frac{R}{2L} I$	

(iii)	Since $R = 4$, $L = 3$ and $C = 0.75$ are all constants, $\therefore \frac{4L}{R^2} = \frac{4(3)}{4^2} = \frac{3}{4} = C,$ we can use (1) from part (ii): $\begin{aligned}\frac{dI}{dt} &= Ae^{\frac{4t}{2(3)}} - \frac{4}{2(3)}I \\ &= Ae^{\frac{2t}{3}} - \frac{2}{3}I \\ &= Ae^{\frac{2t}{3}} - \frac{2}{3}Ate^{\frac{2t}{3}}\end{aligned}$ For stationary point, $\frac{dI}{dt} = 0$ $e^{\frac{2t}{3}} \left(1 - \frac{2}{3}t\right) = 0 \quad (\text{Since } A > 0)$ $t = \frac{3}{2} \quad \left(\text{Since } e^{\frac{2t}{3}} > 0 \text{ for all } t\right)$ $\Rightarrow I = A \left(\frac{3}{2}\right) e^{\frac{4}{2(3)}\left(\frac{3}{2}\right)} = \frac{3A}{2e}$ To show Maximum: Sub $t = \frac{3}{2}$, $I = \frac{3A}{2e}$ and $\frac{dI}{dt} = 0$ into $L \frac{d^2I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = 0$ $3 \frac{d^2I}{dt^2} + 4(0) + \frac{4^2}{4(3)} \left(\frac{3A}{2e}\right) = 0$ $3 \frac{d^2I}{dt^2} = -\frac{2A}{e}$ $\frac{d^2I}{dt^2} = -\frac{2A}{3e} < 0 \quad \left(\text{Maximum at } t = \frac{3}{2}\right)$ Therefore, the maximum value of I is $\frac{3A}{2e}$.	
(iv)	 Label the maximum point found in (iii). Sketch only where t and I are positive. Label the horizontal asymptote	

11 (ia)	Total amount $= \\$100(1.002)^{12}$ $= \\$102.43$ (2 d.p.) Dollars to 2 decimal places.	Interest rate = 0.2% Amt in bank after one month $= \\$100(100\% + 0.2\%)$ $= \\$100(1 + 0.002)$ $= \\$100(1.002)$ 100% denotes the original principal amount in bank. Thus, amount increase by a multiple of 1.002.												
(b)	<table><tr><th>Month</th><th>Start of month</th><th>End of month</th></tr><tr><td>1 Jan</td><td>100</td><td>$100(1.002)$</td></tr><tr><td>1 Feb</td><td>$100(1.002) + 100$</td><td>$100(1.002^2) + 100(1.002)$</td></tr><tr><td>1 Mar</td><td>$100(1.002^2) + 100(1.002) + 100$</td><td>$100(1.002^3) + 100(1.002^2) + 100(1.002)$</td></tr></table> For interest rate question, we need to draw a table and fill in the total amount at the start and end of every month to get a pattern. Note, to get a pattern, we do not simplify the values. Total amount at end of 31 Dec 2016 (12 months) $= 100(1.002)^{12} + 100(1.002)^{11} + \dots + 100(1.002)$ $= 100(1.002) + 100(1.002)^2 + \dots + 100(1.002)^{12}$ $= \frac{100(1.002)}{1 - 1.002} (1 - 1.002^{12})$ $= 1215.71$ $= \\$1215.71$ (2 d.p.) Dollars to 2 decimal places. Always arrange in ascending powers for the multiple for ease of calculation. Sum of GP: $\frac{\text{first term}}{1 - r} (1 - r^{\text{no. of terms}})$, r = common ratio		Month	Start of month	End of month	1 Jan	100	$100(1.002)$	1 Feb	$100(1.002) + 100$	$100(1.002^2) + 100(1.002)$	1 Mar	$100(1.002^2) + 100(1.002) + 100$	$100(1.002^3) + 100(1.002^2) + 100(1.002)$
Month	Start of month	End of month												
1 Jan	100	$100(1.002)$												
1 Feb	$100(1.002) + 100$	$100(1.002^2) + 100(1.002)$												
1 Mar	$100(1.002^2) + 100(1.002) + 100$	$100(1.002^3) + 100(1.002^2) + 100(1.002)$												

(c) Total amount at the end of n months

$$= 100(1.002)^n + 100(1.002)^{n-1} + \dots + 100(1.002)$$

$$= 100(1.002) + 100(1.002)^2 + \dots + 100(1.002)^n$$

$$= \frac{100(1.002)}{1-1.002}(1-1.002^n)$$

$$= 50100((1.002)^n - 1)$$

Always arrange in **ascending** powers for the multiple for ease of calculation.

Sum of GP:

$$\frac{\text{first term}}{1-r}(1-r^{\text{no. of terms}}), \quad r = \text{common ratio}$$

Using GC,

when $n = 29$, total amount = \$2988.67 (31 May 2018)

when $n = 30$, total amount = \$3094.80 (30 Jun 2018)

This only indicates to us that it exceeds \$3000 in June. We still need to check if it is the first or last day of June.

X	Y1
28	2882.7
29	2988.6
30	3094.8
31	3201.2
32	3307.8
33	3414.6
34	3521.7
35	3628.9
36	3736.4
37	3844
38	3951.9

Use Table of Values as n is an integer

On 31 May 2018, the total amount is \$2988.67.

On 1 June 2018, the total amount is \$2988.67 + \$100 = \$3088.67.

From 31 May 2018 to 1 June 2018, Mr Wong deposits another \$100.

Therefore, the total amount first exceeds \$3000 on 1 June 2018.

Must give the date, indicating clearly if it is the start or end of the month.

(iia)

Total amount
= \$(100+12b)

For the first \$100 invested, Mr Wong gets \$b at the end of **every month**. Thus, at the end of 12 months, he will get 12 times of \$b.

(b)

Month	Start of month	End of month
1 Jan	100	100+b
1 Feb	100+b+100	(100+b)+(100+2b)
1 Mar	(100+b)+(100+2b)+100	(100+b)+(100+2b)+(100+3b)

For interest question, we need to draw a table and fill in the total amount at the start and end of every month to get a pattern. Note, to get a pattern, we do not simplify the values.

Total Amount at the end of 24 months = 2800

$$(100 + 24b) + (100 + 23b) + \dots + (100 + b) = 2800$$
$$(100 + b) + (100 + 2b) + \dots + (100 + 24b) = 2800$$
$$\frac{24}{2}((100 + b) + (100 + 24b)) = 2800$$

Always arrange in **ascending order** for ease of calculation.

$$200 + 25b = \frac{700}{3}$$
$$b = 1.33 \text{ (2d.p)}$$

Sum of AP:

$$\frac{\text{no. of terms}}{2}(\text{1st} + \text{last})$$

Dollars to 2 decimal places.

(iii)

Sum of GP:

$$\frac{\text{first term}}{1-r} (1-r^{\text{no. of terms}}), \quad r = \text{common ratio}$$

Sum of AP:

$$\frac{\text{no. of terms}}{2} (\text{1st} + \text{last})$$

The new interest rate is 1%.
So the amount increase by a multiple of 1.01 per month.

$$(100+60b) + (100+59b) + \dots + (100+b) = 100(1.01)^{60} + 100(1.01)^{59} + \dots + 100(1.01)$$

$$(100+b) + (100+2b) + \dots + (100+60b) = 100(1.01) + 100(1.01)^2 + \dots + 100(1.01)^{60}$$

$$\frac{60}{2} ((100+b) + (100+60b)) = \frac{100(1.01)}{1-1.01} (1-1.01^{60})$$

Using GC, $b = 1.23$ (2 d.p.)

Always arrange in **ascending order** for ease of calculation.

Dollars to 2 decimal places.

NORMAL FLOAT AUTO REAL Radian MP

Plot1 Plot2 Plot3

$Y_1 = \frac{60}{2} (100 + X + 100 + 60X)$

$Y_2 = 100(1.01)^X \times \frac{1-1.01^{60}}{1-1.01}$

$Y_3 =$

$Y_4 =$

$Y_5 =$

$Y_6 =$

$Y_7 =$

NORMAL FLOAT AUTO REAL Radian MP

ZOOM MEMORY

9↑ZoomStat

0:ZoomFit

A:ZQuadrant1

B:ZFrac1/2

C:ZFrac1/3

D:ZFrac1/4

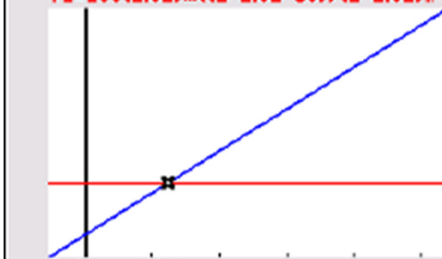
E:ZFrac1/5

F:ZFrac1/8

G:ZFrac1/10

NORMAL FLOAT AUTO REAL Radian MP
CALC INTERSECT

$Y_2 = 100(1.01)^X \times ((1-1.01^{60}) / (1-1.01))$



Select **0:ZoomFit** to get the graphs in an appropriate window.